



CS 561: Map Reduce and Big Data

Course Code	CS 561
Course Name	Map Reduce and Big Data
L-T-P-C	3-0-0-3
Intended for	UG/PG
Distribution	Elective
One-time approval	9th Senate (May 2016)

Course Contents

Modules:

- **Unit 1:** Introduction to Big Data, the MapReduce paradigm and programming model. The MapReduce framework and its benefits. Cloud computing and MapReduce. Open source MapReduce frameworks Hadoop, Shark, Mrjob etc. - comparison and benefits of each such framework. [3 Lectures]
- **Unit 2:** Thinking in the MapReduce way via simple problems that serve as a building block for larger problems Matrix-Vector multiplication, Matrix multiplication, Relational algebra selections, projections, union, intersection, difference, natural join, grouping and aggregation. Complexity analysis of MapReduce algorithms. [3 Lectures]
- **Applying MapReduce to different Big Data contemporary problem areas:** [36 Lectures]
 - Similarity computations
 - Clustering algorithms K-means, CURE, ...
 - Web crawling and indexing
 - Web-scale graph algorithms PageRank, HITS, ...
 - Recommendation Systems content-based filtering, collaborative filtering, dimensionality reduction.
 - Text advertising on the web AdWords.
 - Social networks and their analysis.
 - Large-scale Machine Learning Perceptrons and Support Vector machines.
 - Statistical Machine Translation.
 - Market Basket analysis.
 - ? Mining Data Streams.

Text Books

1. Jimmy Lin and Chris Dyer, **Data-Intensive Text Processing with Map Reduce**, Morgan and Claypool.
2. Rajaraman and Ullman, **Mining of massive Datasets**, Cambridge University Press